

Finding the Structure of Documents

- In human language, words and sentences do not appear randomly but usually have a structure.
- For example, combinations of words form sentences, meaningful grammatical units, such as statements, requests, and commands.
- Automatic extraction of structure of documents helps subsequent NLP tasks like parsing, machine translation, and semantic labelling that use sentences as the basic processing unit.
- **Sentences boundary detection** allows the task of deciding where sentences start and end given a sequence of characters.

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- **Topic segmentation** allows the task of determining when a topic starts and ends in a sequence of sentences.
- The statistical classification approaches that try to find the presence of sentence and topic boundaries given human-annotated training data for segmentation.
- These methods base their predictions on features of the input and local characteristic that give evidence toward the presence or absence of a sentence.
- Features are the core of classification approaches and require careful design and selection in order to be successful and prevent overfitting and noise problem.

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- For example,
- For processing of Chinese documents, the processor may need to first segment the character sequences into words, as the words usually are not separated by a space in Chinese documents.
- Similarly, for morphological rich languages, the word structure may need to be analyzed to extract additional features.
- Such processing is usually done in a pre-processing step, where a sequence of tokens is determined.

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1. Sentence Boundary Detection

- **Sentence boundary detection** (Sentence segmentation) deals with automatically segmenting a sequence of word tokens into sentence units.
- In written text for example in English and in some other languages, the beginning of a sentence is usually marked with an uppercase letter, and the end of a sentence is explicitly marked with a period(.), a question mark(?), an exclamation mark(!), or any other type of punctuation.
- In addition to their role as sentence boundary markers, capitalized initial letters are used to distinguish proper nouns.

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For Example:

- **I spoke with Dr. Smith. My house is on Mountain Dr.**
- In the first sentence, the abbreviation Dr. does not end a sentence, and in the second it does.
- An automatic method that outputs word boundaries as ending sentences according to the presence of such punctuation marks would sometime result in cutting some sentences incorrectly.
- Sentence Boundary Detection (SBD) is a natural language processing (NLP) task that involves identifying the boundaries or boundaries between the sentences in a text.

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- It is a crucial preprocessing step in many NLP applications, as correctly segmenting a text into sentences is necessary for tasks such as text summarization, machine translation, sentiment analysis, and part-of-speech tagging.
- **Objective of Sentence Boundary Detection:**
 1. The main goal of Sentence Boundary Detection is to determine where one sentence ends and the next one begins within a given text.
 2. Sentence boundary detection (Sentence segmentation) deals with automatically segmenting a sequence of word tokens into sentence units.

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- **Challenges in Sentence Boundary Detection:**
- Detecting sentence boundaries can be challenging due to various factors, including:
 1. **Ambiguities:** Sentences can end with different punctuation marks (e.g., ".", "!", "?") or even none at all.
 2. **Abbreviations:** Periods can appear in the middle of sentences within abbreviations, making it hard to distinguish sentence boundaries.
 3. **Domain-Specific Text:** In some domains, like legal or medical texts, sentences may have unconventional structures that are different from typical written language.

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4. Multilingual Text: Different languages may have different sentence boundary conventions, making SBD more complex in multilingual contexts.

Techniques for Sentence Boundary Detection

Several techniques and approaches are used for Sentence Boundary Detection:

- 1. Rule-Based Methods:** These methods rely on a set of rules and heuristics to determine sentence boundaries.
 - Rules may consider punctuation marks, abbreviations, and contextual information.

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2. Machine Learning: Machine learning techniques, such as sequence labeling using Conditional Random Fields (CRF) or Recurrent Neural Networks (RNNs), can be trained to predict sentence boundaries based on labeled training data.

3. Language-Specific Models: Some languages may require language-specific models and resources (e.g., dictionaries, grammatical rules) to perform accurate Sentence Boundary Detection.

4. Pre-trained Models: Pre-trained language models, like BERT or GPT, can be fine-tuned for Sentence Boundary Detection tasks, leveraging their contextual understanding of text.

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- **Topic Boundary Detection** (topic segmentation)
- Topic boundary detection in NLP is the task of identifying where one topic ends and another begins in a text.
- Given a sequence of (written or spoken) words, the aim of topic segmentation is to find the boundaries where topics change.
- Topics can be expressed in a variety of ways, including explicitly (e.g., using topic headings or keywords) or implicitly (e.g., through the use of discourse markers or logical transitions).

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- **Objective of Topic Boundary Detection.**
 1. The primary goal of Topic Boundary Detection is to identify points in a text where the topic or subject matter changes significantly.
 2. These boundaries can be identified at the sentence, paragraph, or document level, depending on the specific application.
- Topic segmentation is an important task for various language understanding applications, such as information retrieval, machine translation and text summarization.

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- **For example**
- In information retrieval, if a long documents can be segmented into shorter, topically coherent segments, then only the segment that is about the users query could be retrieved.
- **Challenges in Topic Boundary Detection**
 1. **Topic ambiguity:** It can be difficult to define what constitutes a topic, and different people may have different interpretations of the same text.
 2. **Language Variations:** Different languages and writing styles may have distinct ways of signaling topic changes.

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- Several techniques and approaches are used for Topic Boundary Detection.
 1. **Rule-Based Methods:** Rule-based methods use predefined rules and heuristics to identify topic boundaries.
 2. **Machine Learning:** Machine learning techniques, such as supervised classification or sequence labeling, can be trained to predict topic boundaries based on labeled training data.
 3. **Topic Models:** Topic modelling techniques like Negative Matrix Factorization (NMF), can be used to identify topic transitions based on changes in topic distributions within a document or corpus.

Methods for Sentence segmentation and Topic segmentation

- Sentence segmentation and topic segmentation have been considered as a **boundary classification problems**.
- Given a boundary candidate(between two word tokens for sentence segmentation and between two sentences for topic segmentation), the goal is to predict whether or not the candidate is an actual boundary (sentence or topic boundary).
- Formally, let $\mathbf{x} \in \mathbf{X}$ be the **vector of features** (the observation) associated with a candidate and $y \in \mathbf{Y}$ be the **label** predicted for that candidate.
- The label y can be **b** for **boundary** and ***b'*** for **non-boundary**.

Methods for Sentence segmentation and topic segmentation

- **Classification problem:** Given a set of training examples $(\mathbf{x}, y)$, find a function that will assign the most accurate possible label y of unseen examples $\mathbf{x}$ which is unseen.
- Alternatively to the binary classification problem, it is possible to model boundary types using finer-grained categories.
- For segmentation the text be framed as a three-class problem: sentence boundary $\mathbf{b}^a$, without an abbreviation $\mathbf{b}^{a'}$ and abbreviation not as a boundary $\mathbf{b}^{-a}$
- Similarly for spoken language, a two way classification can be made between non-boundaries statements $\mathbf{b}_s$, and question boundaries $\mathbf{b}_q$.

Methods for Sentence segmentation and topic segmentation

- For sentence or topic segmentation, the problem is defined as finding the most probable sentence or topic boundaries.
- The natural unit of sentence segmentation is words and of topic segmentation is sentence, as we can assume that topics typically do not change in the middle of a sentences.
- The words or sentences are then grouped into categories belonging to one sentences or topic as boundaries and non-boundaries.
- The classification can be done at each potential boundary i (local modelling); then, the aim is to estimate the most probable boundary type $\hat{y}_i$ for each candidate X_i .

Methods for Sentence segmentation and topic segmentation

$$\hat{y} = \underset{y_i \text{ in } Y}{\operatorname{argmax}} P(y_i | x_i)$$

- Here, the $\hat{}$ is used to denote estimated categories, and a variable without a $\hat{}$ is used to show possible categories.
- It is also possible to see the candidate boundaries as a sequence and search for the sequence of boundary types $\hat{y}_i = \hat{y}_1, \dots, \hat{y}_n$ that have maximum probability given the candidate examples $X = x_1, \dots, x_n$.

$$\hat{Y} = \underset{y}{\operatorname{argmax}} P(Y|X) \text{ ----- 2.1}$$

- We categorize the methods into local and sequence classification.

Methods for Sentence segmentation and topic segmentation

- Another categorization of methods is done according to the type of the machine learning algorithm: **Generative versus Discriminative**.
- **Generative sequence models** estimate the joint distribution of the observations $P(X,Y)$ (words, punctuation) and the labels (sentence boundary, topic boundary).
- **Discriminative sequence models**, however, focus on features that categorize the differences between the labelling of the examples.

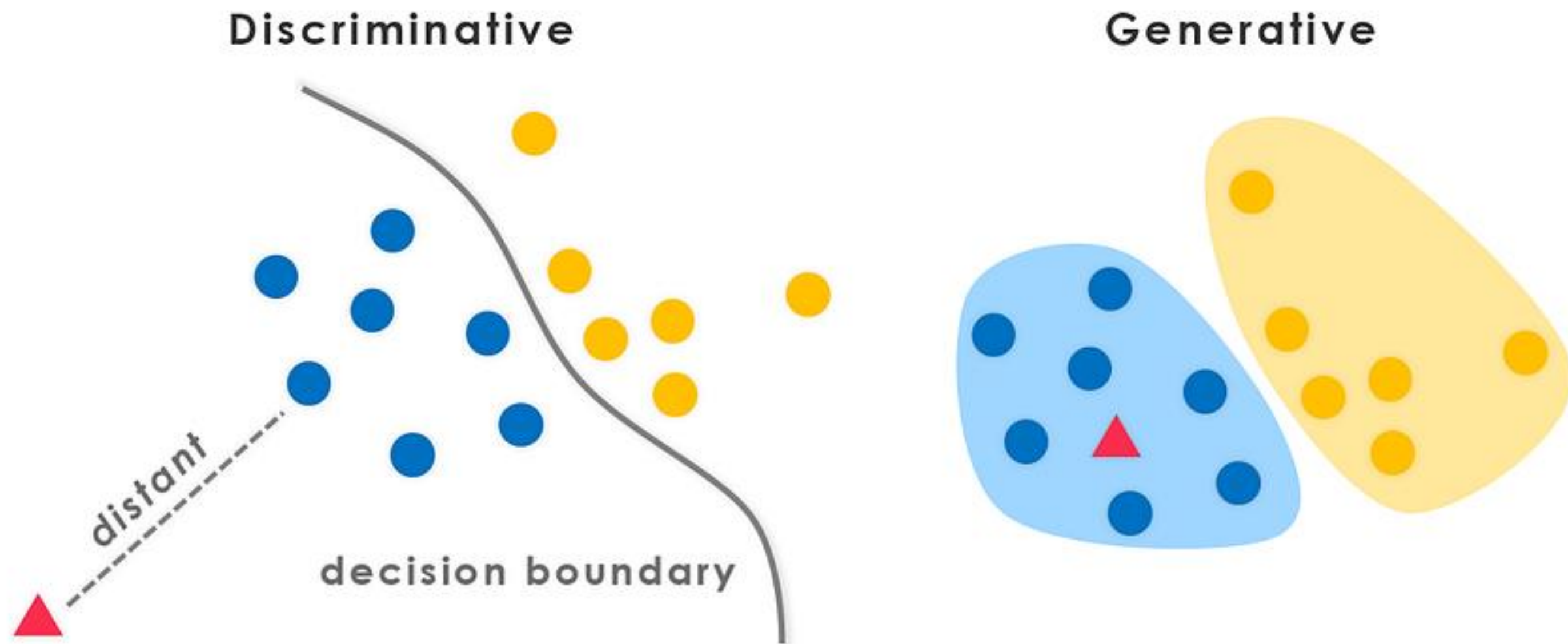
Generative and Discriminative Sequence Models

- NLP encompasses various tasks, such as text classification, sentiment analysis, machine translation, and language generation.
- Several models have been developed to address these tasks, the two primary models in NLP being generative and discriminative models.
- Generative and discriminative sequence models are two categories of statistical models used in natural language processing (NLP) and in various other fields.
- They have different approaches to modeling and solving sequence-related tasks:

Generative and Discriminative Sequence Models

- Discriminative models map an input to output and are trained on labeled data following the supervised learning paradigm.
- These models learn to identify patterns and correlations between the input and output, allowing them to make highly accurate predictions.
- The discriminative model indirectly learns certain features of the dataset that makes the task easier.
- Discriminative models work by learning the decision boundary that separates the input data into different classes.

Generative and Discriminative Sequence Models



- Discriminative modelling learns to model the conditional probability of class label y given set of features x as $P(Y|X)$.
- Generative models are probabilistic models that can generate new text based on the input given to them.

Generative and Discriminative Sequence Models

- These models are trained on large amounts of unlabeled data and can be fine-tuned to perform various NLP tasks.
- They work by learning the probability distribution of words in a language and use this knowledge to generate new text that matches the input's context.
- Generative modelling defines how a dataset is generated.
- It tries to understand the distribution of data points, providing a model of how the data is actually generated in terms of a probabilistic model.

Generative and Discriminative Sequence Models

- The aim is to generate new samples from what has already been distributed in the training data.
- The advantage of using generative model is it helps to represent the data more realistically.
- Generative modelling learns to approximate $P(X)$, which is the probability of observing from the observation X .
- **For Example**
- Let's say you have input data $\mathbf{x}$ and you want to classify the data into labels y .
- A generative model learns the joint probability distribution $P(X,Y)$ and a discriminative model learns the conditional probability distribution $P(Y | X)$.

Generative and Discriminative Sequence Models

- Suppose you have the following data in the form (x,y) : $(1,0)$, $(1,0)$, $(2,0)$, $(2, 1)$.

- $p(x,y)$ is

	$Y = 0$	$Y = 1$
$X = 1$	0.5	0
$X = 2$	0.25	0.25

$p(y | x)$ is

	$Y = 0$	$Y = 1$
$X = 1$	1	0
$X = 2$	0.5	0.5

- The distribution $p(y | x)$ is the natural distribution for classifying a given example x into a class y , which is why algorithms that model this directly are called discriminative algorithms.
- Generative algorithms model $p(x,y)$, which can be transformed into $p(y | x)$ by applying Bayes rule and then used for classification.

Generative and Discriminative Sequence Models

- Generative models are not as simple as discriminative. A generative model describes how likely each topic is, and how likely words are given in the topic.
- This is how it says documents are actually "generated" by the word -- a topic arises according to some distribution, words arise because of the topic, that have a document.
- Classifying documents of words W into topic T is a matter of maximizing the joint likelihood: $P(T, W) = P(W | T)P(T)$.
- A discriminative model operates by only describing how likely a topic is, given the words without knowing how likely the words or topic in a document.

Generative and Discriminative Sequence Models

- The task is to model $P(T | W)$ directly and find the T that maximizes this. The discriminative approaches do not care about $P(T)$ or $P(W)$ directly.
- In discriminative models, to predict the label y from the training example x , we must evaluate:

$$\hat{Y} = \underset{y}{\operatorname{argmax}} P(Y|X)$$

- which merely chooses what is the most likely class y considering x .
- It's like we were trying to model the decision boundary between the classes.
 - Now, using Bayes' rule, let's replace $P(Y | X)$ in the equation by

$$\frac{P(x|y)P(y)}{P(x)}$$

Generative and Discriminative Sequence Models

- We are just interested in the *argmax*, eliminate the denominator that will be the same for every y .

- So, we get :

$$\hat{Y} = \underset{y}{\operatorname{argmax}} (P(X|Y)P(Y))$$

- which is the equation used in generative models.
- In the first case, the conditional probability distribution $P(Y|X)$, which modeled the boundary between classes.
- In the second case, the joint probability distribution $P(X, Y)$, since $P(X | Y) P(Y) = P(X, Y)$, which explicitly models the actual distribution of each class.

Generative and Discriminative Sequence Models

- With the joint probability distribution function, given a Y, calculate ("generate") its respective X.
- For this reason, they are called "generative" models.

$$\hat{Y} = \underset{y}{\operatorname{argmax}} (P(X|Y)P(Y)/P(X)) \text{ ----- 2.2}$$

$$\hat{Y} = \underset{y}{\operatorname{argmax}} (P(X|Y)P(Y)) \text{ ----- 2.3}$$

- Here Y Predicted class(boundary) label
- $Y=(y_1,y_2,\dots,y_k)$ =Set of class (boundary) labels
- $X=(x_1,x_2,\dots,x_n)$ =set of feature vectors
- $P(Y|X)$ the probability of given the X (feature vectors), what is the probability of X belongs to the class(boundary) label $P(x)$

Methods for Sentence segmentation and topic segmentation

- $P(x)$ Probability of word sequence
- $P(Y)$ = Probability of the class(boundary)
- $P(X)$ in the denominator is dropped because it is fixed for different Y and hence does not change the argument of max.
- $P(X | Y)$ and $P(Y)$ can be estimated as
- $P(X|Y) = \prod_{i=1}^n P(x_i|y_1 \dots \dots y_i)$ ----- 2.4
- and
- $P(Y) = \prod_{i=1}^n P(y_i|y_1 \dots \dots y_{i-1})$ ----- 2.5

Methods for Sentence segmentation and topic segmentation

- **Generative vs Discriminative Modeling**

1. Both Generative (G) and Discriminative (D) can be used for classification.
2. G models work with both supervised and unsupervised learning. But D models are only used for supervised learning problems.
3. The goal of the D model is to estimate the conditional probability $P(Y|X)$. In contrast, the G model learns to approximate $P(X)$ and $P(X|Y)$ in an unsupervised setting, then deduces $P(Y|X)$ in a supervised setting.

Methods for Sentence segmentation and topic segmentation

4. D models learn a linear and non-linear decision boundary, using the data points and their respective labels, without knowing how the data was generated. In learning to model the probability distribution of the data, G models get to understand the data's underlying characteristics.
5. In a supervised setting, D models are known to outperform G models. Especially, when G models do not fit the data well.
6. G models can provide rich insights about the data, when you do not have labels.

Complexity of Approaches

- Sentence/topic segmentation approaches can be rated in terms of complexity (time and memory) of their training and prediction algorithms and in terms of their performance on real-world dataset.
- Some approaches may require specific pre-processing, such as converting or normalizing continuous features to discrete features.
- The complexity of the approaches can be seen in
 1. Discriminative approach
 2. Generative models
 3. Discriminative classifiers and in
 4. Sequence approaches

Complexity of Approaches

1. Discriminative approach

- In terms of complexity, training of discriminative approaches is more complex than training of generative ones.
- Since, they require multiple passes over the training data to adjust for their feature weights.

2. Generative models

- Generative models such as HELMs can handle multiple orders of magnitude larger training sets.
- But, they do not cope well with unseen events.

3. Discriminative classifiers

- They allow for a wider variety of features and perform better on smaller training sets.

Complexity of Approaches

- Predicting with discriminative classifiers is also slower, even though the models are relatively simple (linear or log-linear).

4. Sequence approaches

- Compared to local approaches, sequence approaches bring the additional complexity of decoding:
- To find the best sequence of decisions requires evaluating all possible sequences of decisions.

Performance of the Approaches

- The performance of the sentence segmentation approaches can be analyzed in:
 - A. Sentence segmentation in text
 - B. Sentence segmentation in speech
- **A. Sentence segmentation in text**
- For sentence segmentation in text, researchers have reported error rate results on a subset of the Wall Street Journal Corpus of about 27,000 sentences.
- For instance, Mikheev reports that have rule-based system performs at an error rate of 1.41 %.

Performance of the Approaches

- Even though the error rates presented seem low, sentence segmentation is one of the first processing steps for any NLP task, and each error impacts subsequent steps, especially if the resulting sentences are presented to the user.
- **B. Sentence segmentation in speech**
- For sentence segmentation in speech, performance is usually evaluated using:
 - 1. The error rate (ratio of number of errors to the numbers of examples).
 - 2. F1-measure (the harmonic mean of recall and precision).
- Recall is defined as the ratio of the number of correctly returned sentence boundaries to the number of sentence boundaries in the reference annotations.

Performance of the Approaches

- Precision is the ratio of the number of correctly returned sentence boundaries to the number of all automatically estimated sentence boundaries.

Assignment Questions

1. Discuss the challenges and techniques of sentence boundary detection
2. Explain sentence boundary detection methods with an example?
3. What do you mean by topic boundary detection method?
4. Explain briefly different types of Sequence classification methods?
5. Explain different kinds of performance measures in NLP?
6. Explain any two methods for feature vector extraction in NLP with an examples
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